Paper Code : 21312

F- 412

B.C.A. (Third Semester) Examination, 2024-25

(New Course)

Paper - BCA-302

Computer Organization

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Describe in detail constructing of Arithmetic Logic Unit (ALU)

   (b) What is Floating point arithmetic? Explain with example.

2. (a) Explain the difference between cache memory and virtual memory.

   (b) Explain Memory Hierarchy with example.

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3. (a) Explain segmented memory and paged segment memory.

   (b) Differentiate Magnetic Disk and Magnetic Tape.

4. (a) Explain arithmetic operation on binary operation with example.

   (b) Explain Index Addressing Mode with examples..

5. (a) Explain the DMA control with the help of block diagram.

   (b) Divide using division algorithm 10100011 by 1011

6. (a) Subtract 101 - 0.101 with 2's complement method.

   (b) Differentiate serial bus arbitration with parallel bus arbitration.

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7. (a) What is interrupt? Why it is used? explain with example.

(b) Differentiate between RISC and CISC

8. Write short notes on any four of the following.

(i) Static & Dynamic RAM

(ii) Instruction types.

(iii) Programmed I/O

(iv) High speed memories and its uses

(iv) Octal and Hexa decimal Number system.

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21312

F-412

B.C.A. (Third Semester)

Examination, 2023-24

(New Course)

Paper - BCA-302

Computer Organization

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt **all** sections as directed.

Section-A

Note : Attempt any **five** questions. All questions carry equal marks.

1. (a) Describe various number systems with suitable example.

   (b) How addition and subtraction of numbers is performed using signed 2's complement procedure? Explain with proper example.

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7. (a) Describe the concept of programmed initiated I/O mode of data transfer.

   (b) Draw the work diagram of DMA and illustrate its overall working.

8. Write in brief about the following: (any four)

   (i) Construction of ALU

   (ii) Types of instruction

   (iii) EPROMs and EEPROMs

   (iv) Virtual Memory

   (v) Boolean algebra

   (vi) Cycle Skaling concept

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2. (a) Explain the floating points arithmetic in detail.

(b) Multiply (4) and (–5) using booth's algorithm where (–5) is the multiplier and (4) is multiplicand. Show each step very clearly following the algorithm.

3. (a) What are various types of processor organization? Explain each.

(b) Show the evaluation of following arithmetic expression using stack (step-by-step) : 2*3 + 4*5

4. (a) What do you mean by "Addressing Modes"? Describe them in brief.

(b) Differentiate SRAM and DRAM clearly.

5. (a) Draw and explain the memory hierarchy.

(b) Explain the working of "cache-Memory" with well labelled diagram.

6. (a) How paging is different from segmentation?

(b) What is bus arbitration? Explain its types.

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Paper Code : 21312

F-412

B.C.A. (Third Semester)

Examination, March-2022

(New Course)

Paper - BCA-302

Computer Organization

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Explain the concept of gray code with proper example.

   (b) Discuss the procedure of addition & subtraction of numbers using signed 21s complement using one example for each.

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2. (a) Elaborate the construction process of A.L.U.

   (b) Multiply (–4) and (+5) using Booth's algorithm with –4 as multiplicand & +5 as multiplier.

3. (a) How stack is used for information representation? Explain.

   (b) Describe O–address 1–address, 2–address and 3–address type instruction formats with example.

4. (a) What are various types of addressing modes? Write in brief about each mode.

   (b) Differentiate CISC from RISC.

5. (a) Draw a well labelled memory hierarchy & explain the concept of locality of reference.

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(b) Explain the concept of cache memory. What is hit ratio? Suppose there are four memory references that results in a hit and 6 memory reference results in a cache-miss. Calculate the hit ratio.

6. (a) Write in brief about :

(i) PROMs

(ii) Page Faults

(iii) Paging & Segmentation

(b) What are various page replacement policies? Explain each. What do you mean by TLB?

7. (a) How serial bus arbitration is different from parallel bus arbitration? Justify with suitable block diagram.

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(b) Discuss the concept of direct memory Access with a well labelled block diagram. What is cycle stealing?

8. Write short notes on any **four** of the following:

(a) K-MAP method

(b) Interrupts

(c) SRAM and DRAM

(d) IEEE floating point format

(e) Octal number system

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Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21312

F-412

B.C.A. (Third Semester)

Examination, 2021

(New Course)

Paper - BCA-302

COMPUTER ORGANIZATION

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. All questions carry equal marks.

1. (a) Explain the concept of RISC and CISC

   (b) What is set associative memory? Explain with the help of a block diagram. In which situation set associative memory can be effective utilized.

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2. (a) Explain in brief :

   (i) Address Bus

   (ii) Main Memory

   (iii) Cache Memory

   (b) What are single precision and double precision format. Express 42.687 in IEEE Single Precision point representation.

3. (a) Describe Bus Arbitration. Explain four different types of bus arbitration.

   (b) What are the concept of input-output interfacing and I/O processor.

4. (a) What is the concept of Daisy Chaning. Explain DMA in detail.

   (b) Discuss Cache Memories. What is High speed cache? Explain cache Hit and cache Miss.

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5. (a) Draw a block diagram to illustrate the basic organization of a computer System and explain the function of the logic units.

(b) Explain generation of Computers with their characteristics.

6. (a) Describe the concept of Paging and Page Replacement algorithm.

(b) Consider this page reference stream 12321521625631361243.

Page frame size is 3. Calculate page faults using First In First Out. Optimal Replacement, Least Recently Used.

7. (a) Write down Booth's Algorithm with the help of example.

(b) What are different types of Interrupts describe any one with examples.

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8. Write short notes on any **four** of the following:

(a) Processor organization?

(b) Primary and Secondary memory.

(c) Direct and Indirect addressing.

(d) Thrashing.

(e) Multi programming.

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21312

F–412

B.C.A. (IIIrd Semester) Examination, 2019–20

(New Course)

COMPUTER ORGANISATION

Paper-BCA-302

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) Describe in detail constructing of Arithmetic Logic Unit (ALU).

   (b) What do you mean by Floating point arithmetic ? Explain it with an example.

2. (a) Convert the following numbers accordingly :

   (i) $(10111010)_2 = (?)_{10}$

   (ii) $(ABD45)_{16} = (?)_8$

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   (iii) $(10011010)_2 = (?)_{Gray}$

   (iv) $(128)_{10} = (?)_2$

   (v) $(167)_9 = (?)_2$

   (b) How many types of instruction format ? Describe the all field of instruction format with an example.

3. Describe the following addressing mode with an example :

   (i) Implied mode

   (ii) Immediate mode

   (iii) Register indirect mode

   (iv) Autoincrement or autodecrement mode

   (v) Direct addressing mode

   (vi) Indirect addressing mode

4. Explain the following :

   (i) Micro-operation

   (ii) Micro-instruction

   (iii) Micro-program

   (iv) Micro-code

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5. (a) Differentiate between the following memory :

   (i) RAM and ROM

   (ii) PROM and EEPROM

   (b) How many 256 x 8 RAM chips are needed to provide a memory capacity of 4096 bytes ?

6. (a) Differentiate between segmented memory system and paged segment memory.

   (b) Describe the characteristics of cache memory.

7. (a) Describe the need of bus arbitration in system organization.

   (b) Explain the architecture of DMA in detail.

8. Write short notes on any *four* of the following :

   (i) Logical operation

   (ii) Primary memory

   (iii) Static RAM

   (iv) High speed memory

   (v) Reverse polish notation

   (vi) IEEE floating point representation

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21312

F–412

B.C.A. (Third Semester) Examination, 2018

(New Course)

Paper–BCA–302

COMPUTER ORGANISATION

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. How can one perform multiplication of two numbers by using Booth algorithm for multiplication, also perform 5 × 3 by using Booth's Algorithm ?

2. (a) Convert the following numbers accordingly :

   (i) $(12121)_3 = (?)_{10}$

   (ii) $(FA372)_{16} = (?)_8$

   (iii) $(101101)_2 = (?)_{Gray}$

   (iv) $(1231)_{10} = (?)_2$

   (v) $(175)_9 = (?)_2$

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(b) Obtain the 10's complement of the following six-digit decimal numbers :

(i) 123900

(ii) 090657

3. Explain the following :

(a) Micro-operation

(b) Micro-instruction

(c) Micro-program

(d) Micro-code http://www.mjpruonline.com

4. Explain the purpose of addressing mode, also explain various types of addressing modes.

5. (a) Differentiate between :

(i) Static RAM and dynamic RAM

(ii) EPROM and EEROM

(b) How many 128 × 8 RAM chips are needed to provide a memory capacity of 2048 bytes ?

6. (a) An address space is specified by 24 bits and the corresponding memory space by 16 bits :

(i) How many words are there in the address space ?

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(ii) How many words are there in the memory space ?

(b) Explain paged segment memory.

7. (a) Explain bus arbitration in system organization.

(b) Explain DMA controller with a neat diagram.

8. Write short notes any *four* :

(i) Interrupt

(ii) Main memory addressing

(iii) IEEE floating point representation

(iv) One address instruction

(v) Reverse polish notation

(vi) Direct mapping.

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Paper Code : 21312

F–412

B. C. A. (Third Semester)
EXAMINATION, 2017-18
(New Course)

Paper No. BCA–302–N

COMPUTER ORGANISATION

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Simplify the following Boolean Function in product of sums form by mean of a four variable map. Draw the logic diagram with (i) OR-AND gates (ii) NOR gates $F(w, x, y, z) = \Sigma(2, 3, 4, 5, 6, 7, 11, 14, 15)$ 10

   (b) Show that a JK flip-flop can be converted to a Δ flip-flop with an inverter between the J and K inputs. 4

2. Explain Booth's algorithm for multiplication of two numbers. Show the step by step multiplication process of (+14) * (+ 13) by using Booth's algorithm. 14

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3. What do you mean by addressing modes ? Explain the following addressing modes : 4, 10
   (i) Implied mode
   (ii) Immediate mode
   (iii) Autoincrement mode
   (iv) Register indirect mode
   (v) Index Addressing mode

4. What is instruction format ? Evaluate the arithmetic statement : 14

   $$X = (A + B) * (C + D)$$

   by using :
   (i) Three address instructions
   (ii) Two address instructions
   (iii) One address instructions
   (iv) Zero address instructions
   (v) RISC instructions

5. (a) Explain memory hierarchy in a computer a system. 7

   (b) Explain the following : $3\frac{1}{2}$ each
   (i) Static and Dynamic RAMs
   (ii) PROM and EPROM

6. (a) What is hit ratio ? Explain Associative mapping or Direct mapping for cache memory. 7

   (b) What is page fault ? Explain page replacement for a virtual memory system. 7

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7. Explain the following : $3\frac{1}{2}$ each

   (i) Bus arbitration

   (ii) DMA

   (iii) Types of interrupts

   (iv) Programmed I/O

8. Write short notes on any *four* of the following : $3\frac{1}{2}$ each

   (i) Handshaking

   (ii) Processor Organization

   (iii) BCD code

   (iv) MMV

   (v) Program Counter

   (vi) Universal gates

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